

Given:
 $F = 12 \text{ N}$
 No friction

We first determine the system acceleration.

$$\Sigma \text{ masses} = M + 2M + 3M = 6M$$

$$\Sigma F = 3F - 2F = F \quad \text{since the Tension forces are Newton III pairs, and cancel out in the summation of forces.}$$

$$\therefore \Sigma F = ma$$

$$F = (6M)a$$

$$\Rightarrow a = \left(\frac{F}{6M} \right) \text{ m/s}^2 \quad \text{or} \quad \frac{2}{M} \text{ m/s}^2$$

To determine T_1 , we isolate $3M$ and apply Newton's II to its motion.

$$\Sigma F = ma \quad \text{for mass of } 3M$$

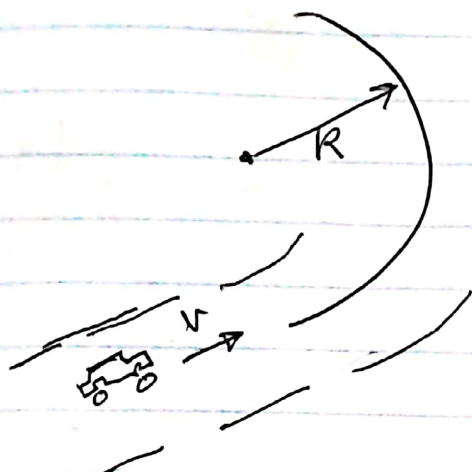
$$T_1 \leftarrow [3M] \rightarrow 3F$$

$$\Rightarrow 3F - T_1 = (3M) \left(\frac{F}{6M} \right)$$

$$3F - T_1 = \frac{F}{2}$$

$$\therefore T_1 = 3F - \frac{F}{2} = (36 - 6) \text{ N} = 30 \text{ N}$$

(5)



$$R = 300 \text{ m}$$

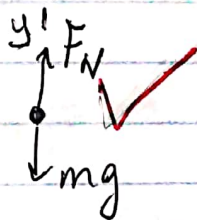
$$v = 120 \text{ km/h}$$

$$= \frac{120\,000 \text{ m}}{1 \text{ h} \times 3600 \text{ s/h}} = 33.33 \text{ m/s}$$

$$\mu_s = 0.250$$

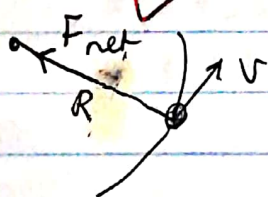
To examine what happens to the car, we need to determine the "sweet" speed at which the vehicle will travel safely through the curve, and compare it with the 120 km/h.

In the vertical direction we have the free-body diagram:



the car is in equilibrium in the vertical direction. $\sum F_y = 0 \Rightarrow F_N = mg$

In the horizontal plane, in the radial direction, the car is under a net force of $\sum F_r$, the centripetal force.



The frictional force f_s provides the centripetal force.

$$\sum F_r = \frac{mv^2}{R}$$

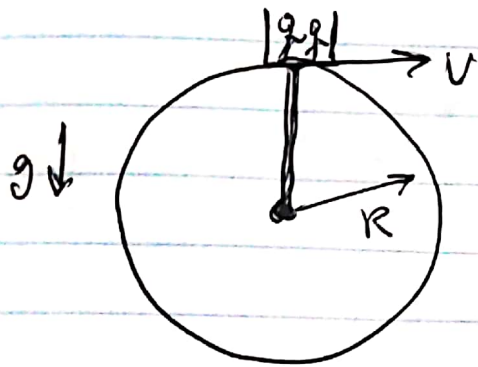
$$f_s = \mu_s F_N = \mu_s mg = \frac{mv^2}{R}$$

$$\therefore v^2 = \mu_s g R$$

$$\Rightarrow v = \sqrt{(0.250)(9.80 \text{ m/s}^2)(300 \text{ m})}$$

$$= 27.1 \text{ m/s}$$

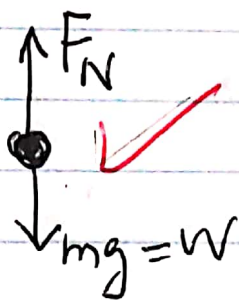
$\therefore$ This speed at which the car enters the curve is greater than the 27.1 m/s, implying veering outside the curve since $f_s < mv^2/R$.



$$R = 10 \text{ m}$$

$$W = mg = 5.0 \text{ kN}$$

$$v = 5.0 \text{ m/s}$$



at the top, $\Sigma F_r = \frac{mv_{\text{top}}^2}{R}$

$$\Rightarrow -F_N + mg = \frac{mv^2}{R}$$

$$\Rightarrow F_N = mg - \frac{mv^2}{R}$$

We can deduce that $m = \frac{W}{g}$

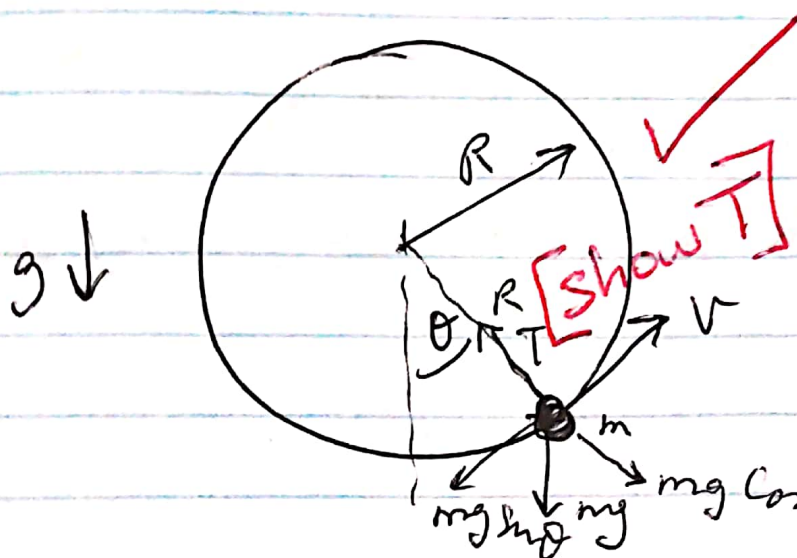
$$\Rightarrow F_N = mg - \frac{mv^2}{R}$$

$$= 5.0 \times 10^3 \text{ N} - \frac{(5.0 \times 10^3 \text{ N})}{(9.80 \text{ m/s}^2)} \frac{(5.0 \text{ m/s})^2}{10 \text{ m}}$$

$$= 3724.48$$

$$= 3.7 \text{ kN} \text{ upwards}$$

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$$m = 0.30 \text{ kg}$$

$$R = 1.6 \text{ m}$$

$$\theta = 50.0^\circ$$

$$T = 8.0 \text{ N}$$

[Resolve mg]

$$\vec{F} = \vec{F}_r + \vec{F}_t$$

$$|\vec{F}| = \sqrt{(T - mg \cos \theta)^2 + (mg \sin \theta)^2}$$

$$= \sqrt{[8.0 \text{ N} - (0.3 \text{ kg})(9.80 \text{ m/s}^2) \cos 50^\circ]^2 - (0.3 \text{ kg})(9.80 \text{ m/s}^2) (\sin 50)}$$

$$= 6.5 \text{ N}$$

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$$F_x = (6.0x - 12) \text{ N}$$

U assigned $+20 \text{ J}$ at $x=0$

$$U_{x_f} - U_{x_i} = - \int_{x_i}^{x_f} F_x dx \quad \checkmark$$

$$= - \int_{x_i}^{x_f} (6.0x - 12) dx$$

$$= - \left\{ \left[\frac{6.0x^2}{2} - 12x \right]_{x_i}^{x_f} \right\} \quad \checkmark$$

$$= - \{ (3.0x_f^2 - 12x_f) - (3.0x_i^2 - 12x_i) \}$$

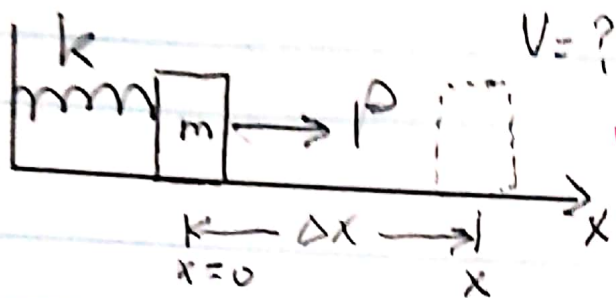
$$= - (3.0)(3.0\text{m})^2 - 12(3.0\text{m}) - 0 \quad \checkmark$$

$$U_{x_f} - 20 \text{ J} = -(27.0 - 36) \text{ J} \quad \checkmark$$

$$= 9.0 \text{ J} + 20 \text{ J}$$

$$\therefore U(x_f) = 29 \text{ J} \quad \checkmark$$

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✓ [Displacement has to be indicated!]

There are no frictional losses.

We take the system to be the spring and the block; there are "other forces" doing work on the system.

$$\therefore W_{\text{other forces}} = \Delta K + \Delta U$$

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$$|P| |\Delta x| \cos \theta = P \Delta x = \left(\frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \right) + \left(\frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2 \right)$$

We can re-write the above in terms of v_f :

$$\frac{1}{2} m v_f^2 = P \Delta x + \frac{1}{2} m v_i^2 - \frac{1}{2} k x_f^2 + \frac{1}{2} k x_i^2$$

$$= P \Delta x + 0 - \frac{1}{2} k x_f^2 + 0$$

$$= \frac{2}{m} \left(P \Delta x - \frac{1}{2} k x_f^2 \right) \text{ or } v_f = \sqrt{\frac{2}{m} \left(P \Delta x - \frac{1}{2} k x_f^2 \right)}$$

$$\therefore v_f = \sqrt{\frac{2}{12 \text{ kg}} \left[(80 \text{ N})(0.13 \text{ m}) - (0.5)(800 \text{ N/m})(0.13 \text{ m})^2 \right]}$$

$$= \frac{1}{6.0 \text{ kg}} (10.4 \text{ Nm} - 6.76 \text{ Nm})$$

$$= 0.78 \text{ m/s}$$

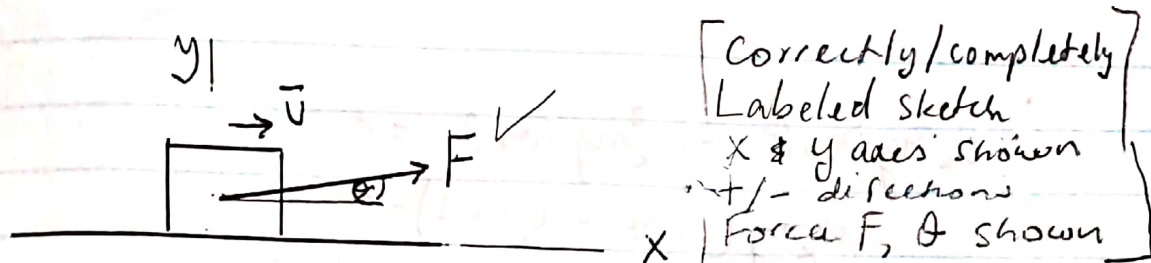
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Next = ΔK should yield the same results also.

$$\vec{P} \cdot \Delta \vec{x} + \vec{F}_s \cdot \Delta \vec{x} = \Delta K$$

$$P \Delta x - \frac{1}{2} k x_f^2 =$$

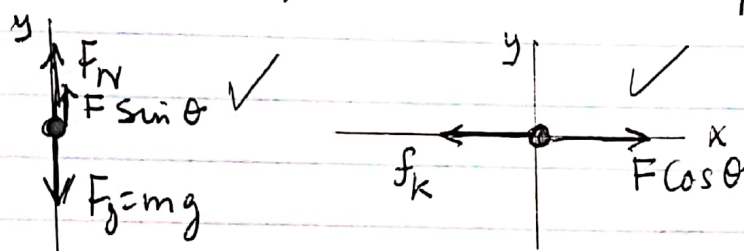
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We apply Newton's II in both the vertical and horizontal axes.

$$\sum F_y = 0 \quad \text{and} \quad \sum F_x = ma_x = 0 \quad \left. \begin{matrix} \text{ } \\ \text{ } \end{matrix} \right\} \text{1 overall}$$

The free-body diagrams in the respective axes:



y-axis: $F_N + F \sin \theta = mg$

$$\therefore F_N = mg - F \sin \theta \quad \text{--- (1)}$$

x-axis: $F \cos \theta - f_k = 0 \Rightarrow F \cos \theta - \mu_k F_N = 0 \quad \text{--- (2)}$

$$F \cos \theta - \mu_k (mg - F \sin \theta) = 0$$

$$F \cos \theta + \mu_k F \sin \theta = \mu_k mg$$

$$\therefore F = \frac{\mu_k mg}{(\cos \theta + \mu_k \sin \theta)}$$

We then minimize: $\frac{dF}{d\theta} = 0$

$$\frac{d}{d\theta} \left[\frac{\mu_k mg}{(\cos \theta + \mu_k \sin \theta)} \right] = 0 \quad \left[\begin{matrix} \text{Substitute correct} \\ \text{expression for F.} \end{matrix} \right]$$

$$\mu_k mg (-1) (\cos \theta + \mu_k \sin \theta)^{-2} \cdot \frac{d}{d\theta} (\cos \theta + \mu_k \sin \theta) = 0 \quad \left[\text{Differentiation} \right]$$

$$\frac{(-\mu_k mg)}{(\cos \theta + \mu_k \sin \theta)^2} (-\sin \theta + \mu_k \cos \theta) = 0$$

$$\therefore \mu_k mg \sin \theta - \mu_k^2 mg \cos \theta = 0$$

$$\mu_k mg (\sin \theta - \mu_k \cos \theta) = 0$$

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$$\therefore \mu_k \cos \theta = \sin \theta \checkmark$$

$$\Rightarrow \mu_k = \tan \theta$$